

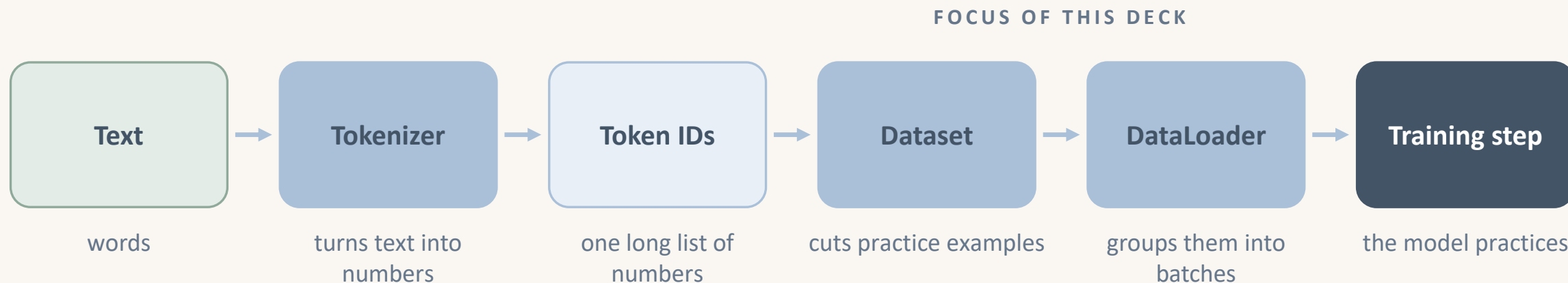
From Tokens to Training Batches

How a dataset and a data loader prepare practice for GPT



The big picture

Before a model can learn from text, the text goes through a few steps.



KEY IDEA

A model learns from many small practice examples, served a few at a time.

Practice: guess the next token

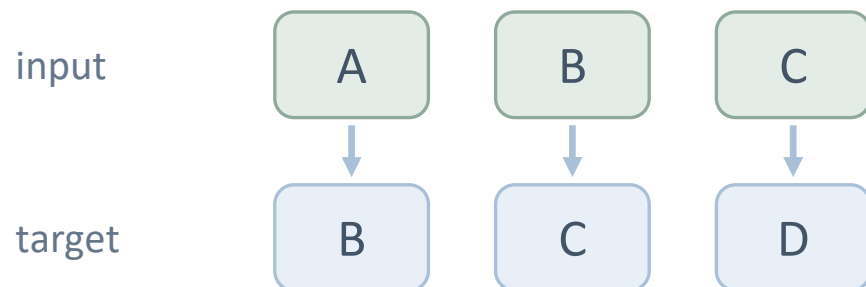
Input = what the model sees. Target = the answers it should guess.

Token stream



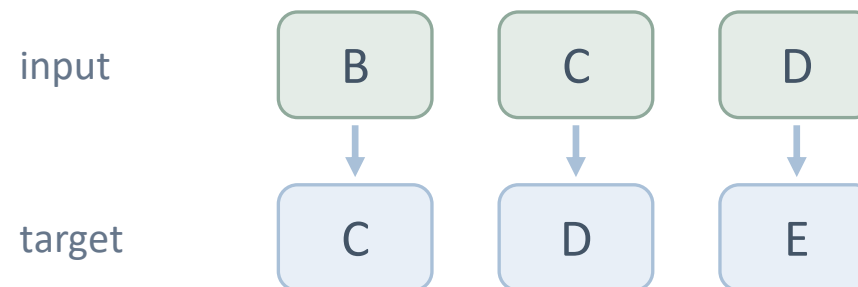
Letters stand in for token IDs.

Example 0



After A comes B. After B comes C.

Example 1



After B comes C. After C comes D.



KEY IDEA

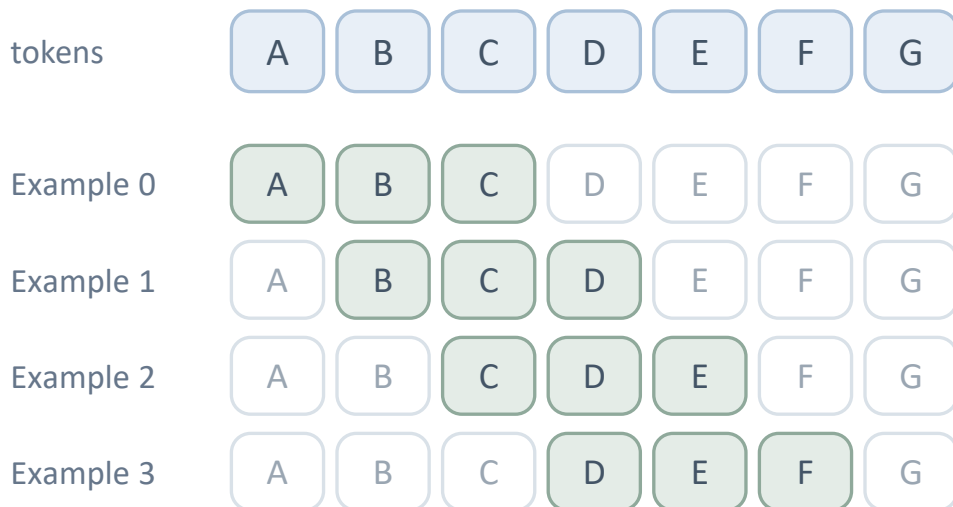
The target is the input moved one token forward.

Sliding a window: max_length and stride

max_length = tokens per example (here, 3). stride = how far the window slides each time.

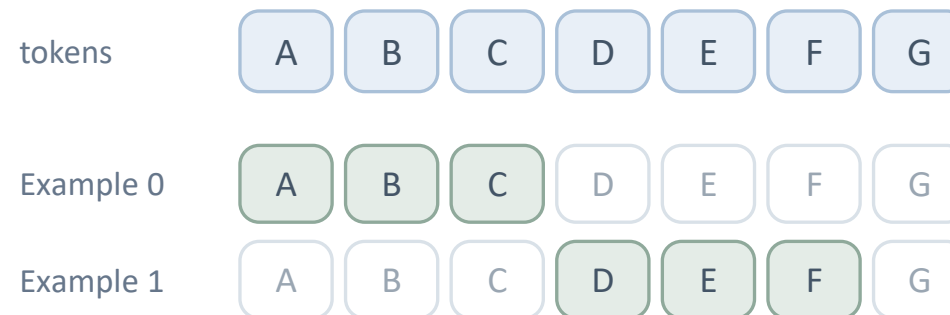
stride = 1

Windows overlap · 4 examples



stride = 3

No overlap · 2 examples



Fewer examples, but each one covers new text.



IN THE CODE

Real settings: max_length 256, stride 128. Each example overlaps the next by half.

The Dataset: a numbered list of examples

A PyTorch Dataset stores practice examples. Each input and target is a 1-D tensor: one row of numbers.

GPT dataset

	input			target		
#0	A	B	C	B	C	D
#1	B	C	D	C	D	E
#2	C	D	E	D	E	F
#3	D	E	F	E	F	G

It answers two questions:

How many examples?

So training knows when it has seen them all.

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Give me example #1.

It returns two 1-D tensors: input and target.

B	C	D
C	D	E



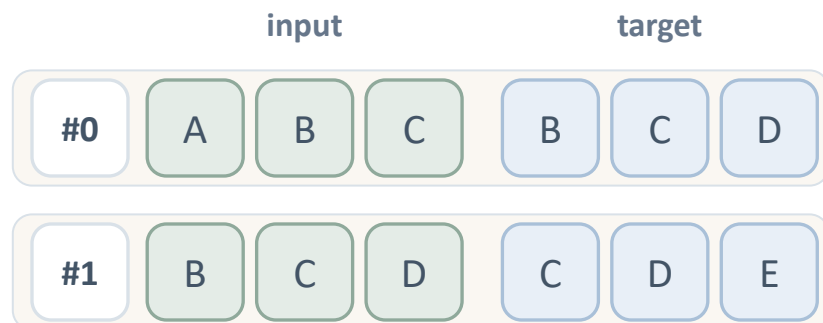
KEY IDEA

Each example is a separate 1-D row. The 2-D grid only appears in a batch.

The DataLoader: examples into batches

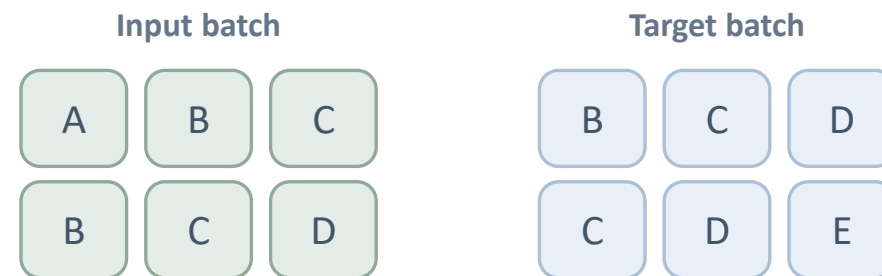
A batch is a group of examples the model practices on in one training step.

Dataset



batch size
= 2

DataLoader



Stacked into 2-D: 2 examples × 3 tokens

Use it: ask by number (#0, #1...)

You get: one example, a 1-D row

Order: the number you ask for

Use it: loop over it, one step at a time

You get: one batch, a 2-D grid

Order: shuffled each pass (by default)

DataLoader settings

batch_size

You choose

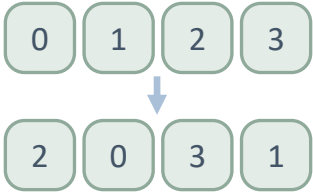
How many examples go in each batch.



shuffle

Default: on

Mixes the example order on each pass through the data.



drop_last

Default: on

Skips a last batch that isn't full, so all batches match.



num_workers

Default: 0

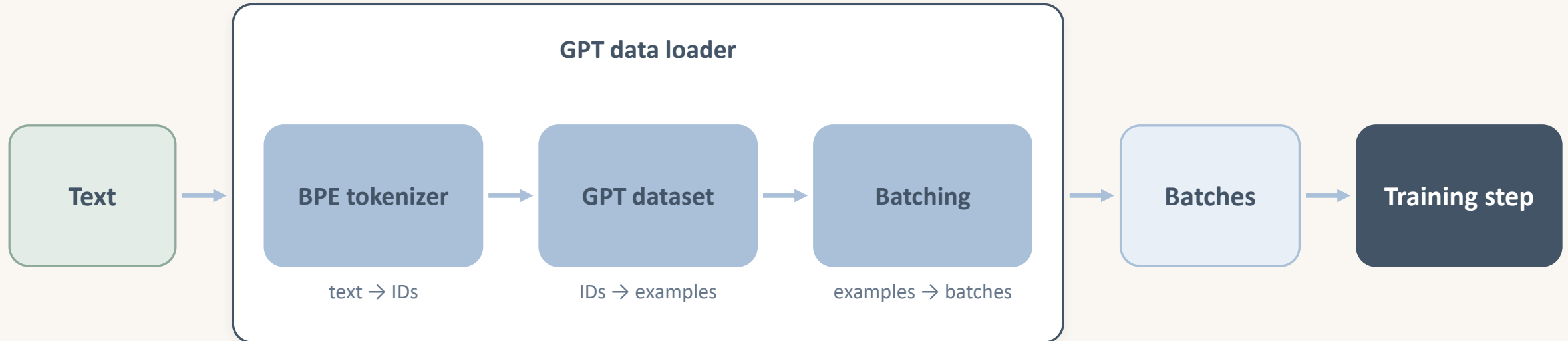
Helpers that load data in the background. 0 = no extra helpers.



Shuffling stops the model from leaning on the order of the text. Same-size batches keep every training step alike.

Putting it together

The GPT data loader sets everything up. You give it text and a batch size.



Each step, the model reads an input batch and checks its guesses against the target batch.



TAKEAWAY

The dataset makes the practice examples. The data loader serves them in batches.